MIT-WPU

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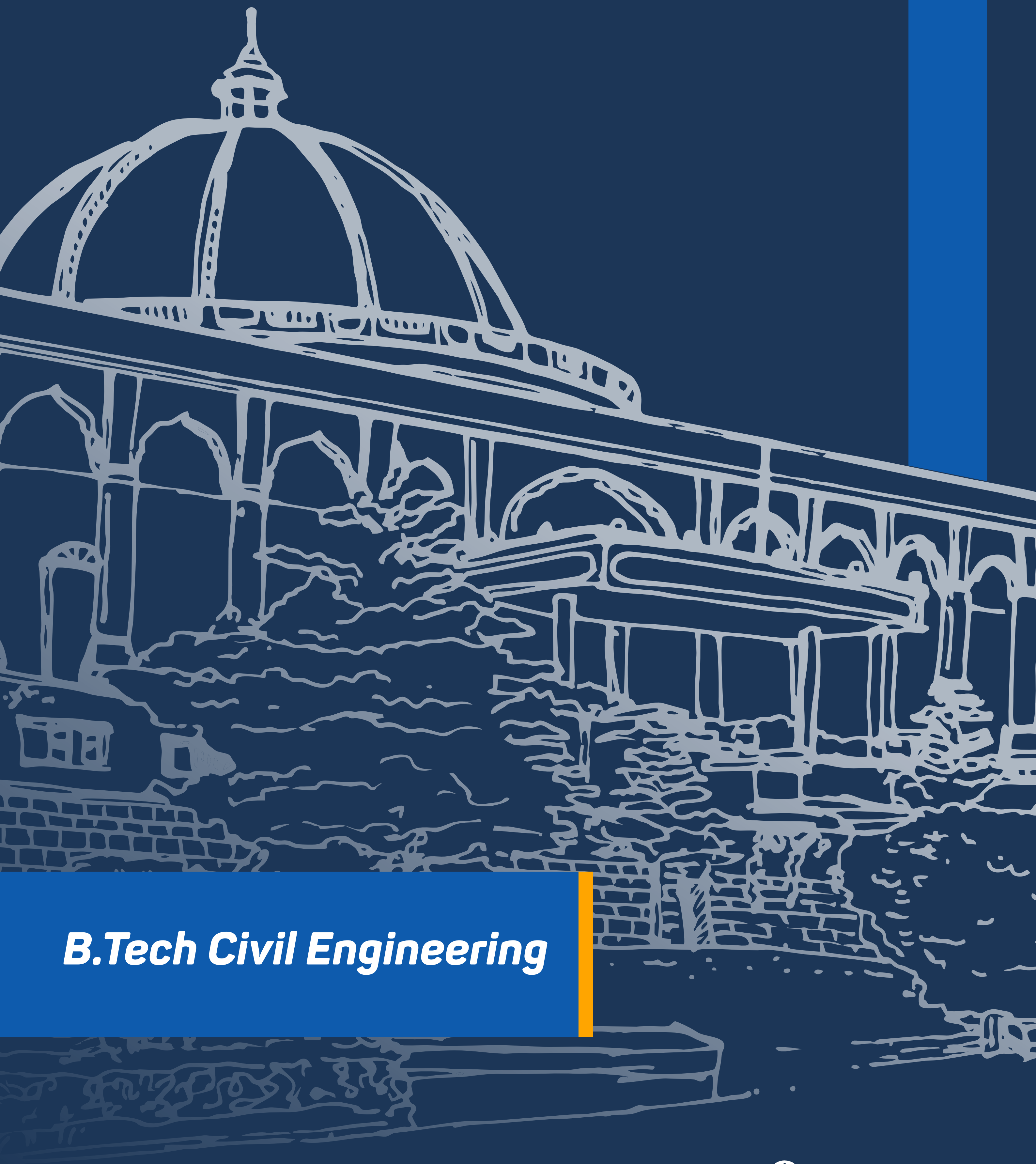
FACULTY OF ENGINEERING AND TECHNOLOGY



B.Tech Civil Engineering

Programme Structure

mitwpu.edu.in

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Civil Engineering
Programme Name	B.Tech. Civil Engineering

Semester	Odd (I)	Even (II)	Total Credits
First Year	22	22	44
Second Year	22	22	44
Third Year	22	22	44
Fourth Year	18	15	33

COURSE BASKET

Course Type	Description
Programme Core	Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU
Programme Electives	Open electives under the programme allow students to specialise in a particular area connected to their major.
University Core	Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students.
University Electives	Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core.

Semester I

Semester	Course Type	Course Name / Course Title	Total Credits
I	UC	Indian Constitution	1
I	UC	Environment and Sustainability	1
I	UC	Yoga - I	1
I	UC	Social Leadership Development Program	1
I	UC	Financial Literacy	1
I	PF	Linear Algebra and Differential Calculus	3
I	PF	Engineering Physics	3
I	PF	Engineering Chemistry	3
I	PF	Engineering Graphics	3
I	PF	Ideas and Innovation in Manufacturing	1
I	PM	Engineering Mechanics	4
		Total Credits:	**22**

Semester II

Semester	Course Type	Course Name / Course Title	Total Credits
II	UC	Yoga - II	1
II	UC	Co-creation	1
II	UC	AI for everyone	2
II	UC	Foundation of Peace	2
II	UC	Indian Knowledge System (General)	2
II	UC	Sports	1
II	PF	Integral Calculus	3
II	PF	Biology for Engineers	2
II	PF	Programming and ProblemSolving (linux based).	3
II	PF	Basic Electrical and Electronics Engineering	3
II	PM	Surveying Field Practices	2
		Total Credits:	**22**

Semester - III

Semester	Course Type	Course Name / Course Title	Total Credits
III	University Core	Spiritual and Cultural heritage: Indian Experience	2
III	University Core	Research Innovation Design Entrepreneurship (RIDE)	1
III	University Electives	University Electives - I	3
III	Program Foundation	Calculus and Numerical methods	4
III	Program Major	Strength of Material	3
III	Program Major	Fluid Mechanics	3
III	Program Foundation	IOT for Smart and Intelligent Buildings	1
III	Program Capstone Project/ Seminar and Internships	Design Thinking	2
III	Program Major	Concrete Technology	3
		Total Credits:	**22**

SEMESTER - IV

Semester	Course Type	Course Name / Course Title	Total Credits
IV	UC	Rural Immersion	1
IV	UC	Life Transformation Skills	1
IV	UE	University Electives - II	3
IV	PF	Probability and Statistics for Civil Engineers	4
IV	PM	Surveying and Geomatic Engineering	3
IV	PM	Structural Analysis I	3
IV	PF	Indian Knowledge System (SCI & Tech.)	2
IV	PM	Soil Mechanics	3
IV	Program Capstone Project/ Seminar and Internships	Data Science for Civil Engineers	2
		Total Credits:	**22**

Semester - V

Semester	Course Type	Course Name / Course Title	Total Credits
V	UC	Managing Conflicts Peacefully: Tools and Techniques	2
V	UE	University Electives - III	3
V	PE	Program Elective - I	4
V	PM	Design of Steel Structures	3
V	PM	Project Management	3
V	Program Capstone Project/Seminar and Internships	Civil Engineering Software	1
V	PM	Structural Analysis II	3
V	PM	Environmental Engineering: Water Engineering	3
		Total Credits:	**22**

Semester - VI

Semester	Course Type	Course Name / Course Title	Total Credits
VI	UC	National Academic Immersion Program	2
VI	PE	Program Elective - II	4
VI	PM	Environmental Engineering: Wastewter Engineering.	3
VI	Program Capstone Project/Seminar and Internships	Building Planning & CAD	3
VI	PM	Design of Reinforced Concrete Structure	3
VI	Program Capstone Project/Seminar and Internships	Hydrology and Irrigation Engineering	3
VI	Program Capstone Project/Seminar and Internships	AIML in Infrastructure Engineering	1
VI	Program Capstone Project/Seminar and Internships	Mini Project	1
VI	Program Capstone Project/Seminar and Internships	BIM in Civil Engineering	2
		Total Credits:	**22**

Semester - VII

Semester	Course Type	Course Name / Course Title	Total Credits
VII	PE	Program Elective - III	4
VII	PF	Transportation Engineering	3
VII	PM	Dams and Hydraulics Structures	3
VII	PM	Quantity Surveying and Estimation	3
VII	Program Capstone Project/Seminar and Internships	Computer aided prestressed concrete design	3
VII	Program Capstone Project/Seminar and Internships	Capstone Project	2
		Total Credits:	**18**

Semester - VIII

Semester	Course Type	Course Name / Course Title	Total Credits
VIII	PE	Program Elective - IV	4
VIII	PF	Foundation Engineering	3
VIII	Program Capstone Project/ Seminar and Internships	AI in Civil Engineering	2
VIII	Program Capstone Project/ Seminar and Internships	Internship	6
		Total Credits:	**15**

Program Electives

Semester	Course Type	Course Name / Course Title	Total Credits
V	PE - I	Advanced Concrete Technology	4
V	PE - I	Structural Audit and Retrofitting	4
V	PE - I	Finite Element Methods	4
V	PE - I	Building Control Procedures and Legislation	4
V	PE - I	Construction Equipment	4
V	PE - I	Tenders and Contracts	4
V	PE - I	Solid Waste Management	4
V	PE - I	Air Pollution and Control	4
V	PE - I	GIS and Remote Sensing	4
V	PE - I	Advanced Surveying and setting out works	4
VI	PE - II	Matrix Methods	4
VI	PE - II	Formwork design and enabling structures	4
VI	PE - II	Earthquake Engineering	4
VI	PE - II	Fastening systems in concrete construction	4
VI	PE - II	Human Resource and Financial Management	4
VI	PE - II	TQM and MIS	4
VI	PE - II	Public Health Engineering	4
VI	PE - II	Architecture and Town Planning	4
VII	PE - III	Advanced Design of Structures	4
VII	PE - III	Ferrocement Design and Technology	4
VII	PE - III	Construction claims and dispute resolutions	4
VII	PE - III	Operational Research	4
VII	PE - III	Environmental assessment methods and legislations	4
VII	PE - III	Advanced Wastewater systems	4
VII	PE - III	Tunnel Engineering	4
VIII	PE - IV	Advanced Geotechnical Engineering	4
VIII	PE - IV	Advanced Foundation Engineering	4
VIII	PE - IV	Real Estate Valuation	4
VIII	PE - IV	Construction Management	4
VIII	PE - IV	Smart Materials in Engineering	4
VIII	PE - IV	Sustainable field Practices	4
VIII	PE - IV	Docks and Harbors	4
VIII	PE - IV	Subsea Engineering	4

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.